NGO CHI NGHIA

AI Engineer
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Summary

AI Engineer specializing in RAG, AI Agent, NLP, and Computer Vision. Co-author of an accepted EAI AICON 2026 paper on energy-based retrieval clustering; 1st in NLP and 4th in Image Classification at the on-site final of FPTU Can Tho Olympic AI (15 finalists from 100+ participants).

Education

Nam Can Tho University, Can Tho, Vietnam. Bachelor of Engineering in Information Technology, GPA 3.44/4.00. Graduated Aug 2026

Experience

AI Engineer (Generative AI), ADHIGHTECH (Tien Phong Engineering Technology Co., Ltd.), Can Tho, Vietnam.Mar 2026 - Present

- Raised answer quality (ROUGE-2 F1 0.49 → 0.68, MRR 0.62, NDCG@5 0.83) on a 120K-chunk Vietnamese economic QA corpus by designing energy-distance cluster ranking with LLM relevance filtering.
- Deployed the QA pipeline as a FastAPI REST service with Docker Compose on Ubuntu.
- Co-authored ERC-RAG, accepted at EAI AICON 2026; contributed to idea development, implementation and experiments.

Publications & Awards

- Co-author, “ERC-RAG: An Energy-based Retrieval Clustering for RAG in Vietnamese Economy Chatbots.” **Accepted at EAI AICON 2026** (camera-ready in preparation).
- FPTU Can Tho Olympic AI, May 2026 (individual, on-site final, 15 finalists from 100+ participants)
  - 1st place, Aspect-Based Sentiment Analysis (NLP) on the private leaderboard. Leaderboard
  - 4th place, Rice Variety Image Classification on the private leaderboard. Leaderboard

Selected Projects

Agent Newsroom: Automated Football News to Vertical Video | GitHubJul - Oct 2026

- Built an end-to-end agent that turns football RSS feeds into vertical videos: story clustering, source cross-checking with a local LLM (Ollama, Qwen3), script generation, FFmpeg rendering, and YouTube publishing.
- Added content safeguards: allowlisted HTTPS sources, article text treated as untrusted input, an independent LLM reviewer, and downgrading stories to “uncertain” when sources conflict.
- Developed a FastAPI + React review interface with per-stage latency telemetry (mean, p50, p95).

PolyAqua Agent: AI Assistant for Shrimp–Crab Polyculture | GitHubApr - May 2026

- Built a FastAPI service for shrimp disease detection from images using YOLOv8n-cls (4 classes), achieving 84% validation and 78% test accuracy.
- Developed a hybrid RAG pipeline (multilingual-E5, BM25, RRF, ChromaDB; 4,593 vectors from 13 technical documents) to generate treatment plans from diagnosis results and pond parameters.
- Implemented calculators for lime dosage, probiotics, and stocking density; added pytest tests and containerized with Docker Compose.

ERC-RAG: Energy-Based Retrieval Clustering for RAG | GitHubMar - May 2026

- Designed a two-stage retrieval framework: top-40 candidates via multilingual-E5 and cosine similarity, K-Means clustering with silhouette-based selection, energy-distance cluster ranking, then LLM relevance filtering before answer generation.
- Evaluated on VNEIQAD (1,000 auto-generated QA pairs over 120,165 VEID chunks) with GPT-5.0 mini: MRR 0.62, NDCG@5 0.83, ROUGE-2 F1 0.68, BLEU-2 0.65.
- Outperformed the group’s prior energy-based reranking baseline on BLEU-2 (0.65 vs 0.61) and ROUGE-2 F1 (0.68 vs 0.49); baseline used GPT-4o-mini.

Skills

Languages: Python, JavaScript, C/C++
Backend: FastAPI, Pydantic, REST API, JWT, OAuth 2.0
AI / ML: PyTorch, TensorFlow, YOLOv8, Hugging Face Transformers, RAG, LangChain, LangGraph, scikit-learn, Sentence Transformers, Prompt Engineering
LLM APIs: OpenAI, Google Gemini, Groq, Ollama
Vector Search: ChromaDB, BM25, Hybrid Retrieval (RRF)
Database: SQLite, PostgreSQL
DevOps & Tools: Docker, Docker Compose, Git, Claude Code CLI, OpenAI Codex CLI
Spoken Languages: Vietnamese (Native), English (VSTEP B1)